STEP 1 — Inventory (230)

UCM			Atmo raw	Atmo average	LYS1 raw	LYS2 raw	LYS average	GVI imagery	SVF imagery	Sound			Solar	Empatica		Eye-tracker	Questionnaire	Metadata / QC				
AIR RH AIR temp HUMIDEX MRT PET SND dBA SUN Gh UTCI WIND AWS AH AIR T bot AIR T mid AIR T top GPS speed kmh	HUMIDEX SR IO flag IR back IR down IR front IR left IR right IR spot gnd IR spot left IR spot right IR spot sky IR up MRT S OPT	PET SR SUN alt SUN az UTCI SR WIND AWA WIND TWD WIND TWS fp	left pm1 left pm10 left pm2.5 right pm1 right pm10 right pm2.5 left humidity left temperature right humidity right temperature	pm1 avg pm10 avg pm2.5 avg	LYS1 kelvin LYS1 lux LYS1 lux cat LYS1 b' LYS1 g' LYS1 medi LYS1 movement LYS1 r' LYS1 rgbb LYS1 rgbg LYS1 rgbir LYS1 rgbr	LYS2 kelvin LYS2 lux LYS2 lux cat LYS2 b' LYS2 g' LYS2 medi LYS2 movement LYS2 r' LYS2 rgbb LYS2 rgbg LYS2 rgbir LYS2 rgbr	kelvin avg lux avg medi avg	GVI 50m GVI grass 50m GVI phase mean GVI plant 50m GVI tree 50m Grass phase mean Plant phase mean Tree phase mean	SVF Building % Ceiling % Combined Vegetation % Grass % Plant % Sky % Tree % Wall %	band proportion 1000-4000 Hz band proportion 125-1000 Hz band proportion 20-125 Hz band proportion 4000-8000 Hz band proportion 8000-20000 Hz category dominant frequency (Hz) mechanisms prop mechanisms score octave proportion 1000 Hz octave proportion 125 Hz octave proportion 2000 Hz octave proportion 250 Hz octave proportion 31.5 Hz	octave proportion 4000 Hz octave proportion 500 Hz octave proportion 63 Hz octave proportion 8000 Hz rural natural prop rural natural score spectral centroid (Hz) spectral rolloff 85% (Hz) unidentify prop urban manmade negative prop urban manmade negative score urban manmade neutral prop urban manmade neutral score urban manmade positive prop	urban manmade positive score vehicle prop vehicle score wind prop wind score	Clearness Index Clearness Index G0 Clearness Index Infer	acc x acc y acc z eda phasic eda scl usiemens eda tonic heart rate hrv fd hf hrv fd hfnu hrv fd lf hrv fd lf hf ratio hrv fd lfnu hrv fd total power hrv fd vlf	hrv td rmssd hrv td sdnn met skin temperature vector magnitude	blink duration s distance from center eyelid aperture avg fixation duration s fixation rate gaze centrality gaze dispersion gaze velocity pupil change rate pupil diameter avg saccade amplitude saccade duration s saccade peak velocity stress composite	current comfort 1 7 environmental comfort mean 1 5 pleasantness 1 5 prs11 being away mean 1 5 prs11 coherence mean 1 5 prs11 fascination mean 1 5 prs11 scope mean 1 5 prs11 total mean 1 5 space familiarity 1 5 stai6 mean 1 5 thermal comfort 1 5	Date Date (Atmo/LYS) Date (UCM) Date (eye) Datetime GPS HDOP GPS alt GPS hdg GPS lat GPS lon GPS speed valid sample count GVI match dist m HRV 5min available HRV 5min max gap (s)	HRV 5min removed pct HRV 5min segment end HRV 5min segment start HRV 5min valid rri count HRV 5min valid span (s) MAG hdg ParticipantID PhaseID SVF Aggregation Method SVF Frames In Window SVF Selection Method SVF Status age bike frequency	bmi category who bmi kg m2 clothing accessories clothing footwear clothing height raw clothing lower body clothing skin tone clothing upper body clothing weight raw commute modes daily walking end time s gender has children	height m nationality own bike for experiment participant id pct complete phase start offset s regular physical activity residence type residential area response index segment site familiarity 1 10 vision correction weight kg	years in brussels

Parameters and decisions by source

The same reason labels are used throughout. “Derived” describes how a measurement is produced; it is not, by itself, a reason to exclude it.

Reason label	Meaning
Aggregation	Raw measurements are replaced by their declared average.
Representation choice	A selected formulation, continuous measure, or derived summary is used instead of an alternative or its constituent inputs; “derived” describes if the parameter is produced from other measurements, and is excluded from the parameter list.
Data availability	Missing observations limit coverage or estimation.
Measurement suitability	Definition, calibration or statistical coding needs verification or a different treatment.
PCA-informed selection	Inventory 34 sound fields → exclude category text and unidentified proportion → standardize 32 numeric exposure inputs → 10 PCs for B4 (k = max(80% rule, 10); 84.04% cumulative) → 10 original-variable candidates → cumulative sound-only VIF by PC order; full set max VIF 2.11; no VIF removals.
Model scope	A valid construct is outside the current equation specification.
Variable role	The field is an outcome, participant descriptor, grouping variable, processing field or QC indicator rather than an environmental predictor.

1. UCM

- Retained: AIR RH, AIR temp, HUMIDEX, MRT, PET, SND dBA, SUN Gh, UTCI, WIND AWS.
- AH: Representation choice — AH is absolute humidity (derived/alternate formulation). ucm_AIR_RH is the declared humidity representative.
- AIR T bot, AIR T mid, AIR T top: Representation choice — ucm_AIR_temp is the air-temperature representative.
- GPS speed kmh: Model scope — Movement measures are excluded.
- HUMIDEX SR, MRT S, PET SR, UTCI SR: Representation choice — Derived alternate from Humidex, MRT, PET, UTCI respectively.
- IO flag: Model scope — IO flag is excluded.
- IR back, IR down, IR front, IR left, IR right, IR spot gnd, IR spot left, IR spot right, IR spot sky, IR up: Representation choice — Directional IR channels feed ucm_MRT (CityFeel17).
- OPT: Representation choice — Derived operative temperature from air temperature and MRT (CityFeel17).
- SUN alt, SUN az: Model scope — Solar altitude and azimuth are excluded.
- WIND AWA, WIND TWD: Representation choice — Derived wind direction/angle and True Wind Direction from ucm_WIND_AWS.
- WIND TWS: Representation choice — Derived true-wind speed from ucm_WIND_AWS.
- fp: Model scope — Proximity score (fp) is excluded.

2. Atmo raw

- left pm1, left pm10, left pm2.5, right pm1, right pm10, right pm2.5: Aggregation — Replaced by the corresponding left/right PM averages. Raw channels are not entered alongside those averages.
- left humidity, right humidity: Representation choice — UCM ucm_AIR_RH is used instead for its higher temporal resolution.
- left temperature, right temperature: Representation choice — UCM ucm_AIR_temp is used instead for its higher temporal resolution.

3. Atmo average

- Retained: pm1 avg, pm10 avg, pm2.5 avg.

4. LYS1 raw

- LYS1 kelvin, LYS1 lux: Aggregation — Combined both LYS1 and LYS2 to form kelvin avg and lux avg.

- LYS1 lux cat: Representation choice — Derived illuminance category.
- LYS1 b', LYS1 g', LYS1 r', LYS1 rgbb, LYS1 rgbg, LYS1 rgbir, LYS1 rgbr: Measurement suitability — Exported RGB/IR or r'/g'/b' channel with no declared average or calibrated H4 exposure.
- LYS1 medi: Derived from LYS1 RGB channels, not a direct spectral measurement.
- LYS1 movement: Model scope — LYS on-body movement signal, not an environmental predictor.

5. LYS2 raw

- LYS2 kelvin, LYS2 lux: Aggregation — Combined both LYS1 and LYS2 to form kelvin avg and lux avg.
- LYS2 lux cat: Representation choice — Derived illuminance category. Continuous lux is available and no averaged replacement exists.
- LYS2 b', LYS2 g', LYS2 r', LYS2 rgbb, LYS2 rgbg, LYS2 rgbir, LYS2 rgbr: Measurement suitability — Exported RGB/IR or r'/g'/b' channel with no declared average or calibrated H4 exposure.
- LYS2 medi: Derived from LYS2 RGB channels, not a direct spectral measurement.
- LYS2 movement: Model scope — LYS on-body movement signal, not an environmental predictor.

6. LYS average

- Retained: kelvin avg, lux avg.
- medi avg: Derived from LYS RGB channels, not a direct spectral measurement.

7. GVI imagery

- Retained: GVI 50m.
- GVI grass 50m, GVI phase mean, GVI plant 50m, GVI tree 50m, Grass phase mean, Plant phase mean, Tree phase mean: Representation choice — Alternate GVI scales, class components and phase summaries from the same vision pipeline.

8. SVF imagery

- Retained: SVF.
- Building %, Ceiling %, Combined Vegetation %, Grass %, Plant %, Sky %, Tree %, Wall %: Sky openness is SVF_Holmer. Scene percentages are separate constructs, not equivalent exposures.

9. Sound

- PCA-informed selection: Within-participant-phase centering and unit-SD scaling precede PCA. B4 selection uses 10 PCs (84.04% cumulative variance; the 80% rule alone would retain 9 PCs at 81.15%). A B4-style sequential rule selects 10 distinct original-variable candidates (PC10: `dominant_frequency_hz`).
- VIF screening: All 10 candidates together have sound-only maximum VIF 2.11. Full Eq1 plus these 10 main effects has maximum VIF 2.884–3.092 across 26 outcome-specific designs; all designs are full rank. No terms are removed.
- freq_prop_20_125, vehicle_prop, wind_prop, freq_prop_8000_20000, mechanisms_prop, octave_prop_125_0_hz, urban_manmade_neutral_prop, freq_prop_4000_8000, urban_manmade_positive_prop, dominant_frequency_hz: Additional main effects in proposed Eq2 (sound sensitivity).
- Interpretation: The 10 original columns do not inherit the PCs' 84.04% variance. This is a proposed sensitivity specification; fitted H4 models have not been updated.

Method, references, selection table and VIF iterations: Sound_screening_pca_vif.md. PCA references: King & Jackson (1999), Jolliffe & Cadima (2016). Regression collinearity: Dormann et al. (2013).

- SND dBA: Retained in Eq1 and included among 32 numeric exposure inputs. Inventory all 34 sound fields first; exclude category text (nominal) and unidentify_prop (classification diagnostic) before PCA.
- Other numeric sound variables: Not selected by the PCA candidate rule. unidentify_prop is excluded before PCA as a classification diagnostic. These decisions do not establish that omitted variables are scientifically unimportant.

10. Solar

- Clearness Index: Derived from Gh/G0 (ClearCI p.3 eq.2).
- Clearness Index G0: Derived from date, time and location only, not personal exposure. Denominator for kt = Gh/G0 (ClearCI p.3 eq.2).
- Clearness Index Infer: Derived categorical clearness from Clearness_Index, not a continuous exposure. No continuous-effect parameterization is declared.

11. Empatica

- acc x, acc y, acc z, met, vector magnitude: Model scope — Movement measures are excluded.

- eda phasic, eda tonic, heart rate, hrv fd hf, hrv fd hfnu, hrv fd lf, hrv fd lf hf ratio, hrv fd lfnu, hrv fd total power, hrv fd vlf, hrv td rmssd, hrv td sdnn, skin temperature: Variable role — Used as a physiological or ocular response, not an environmental predictor.
- eda scl usiemens: Used to compute eda phasic and tonic.

12. Eye-tracker

- blink duration s, distance from center, eyelid aperture avg, fixation duration s, fixation rate, gaze centrality, gaze dispersion, gaze velocity, pupil change rate, pupil diameter avg, saccade amplitude, saccade duration s, saccade peak velocity: Variable role — Used as a physiological or ocular response, not an environmental predictor.
- stress composite: Derived from other eye-tracker fixation measures.

13. Questionnaire

- current comfort 1 7, environmental comfort mean 1 5, pleasantness 1 5, prs11 being away mean 1 5, prs11 coherence mean 1 5, prs11 fascination mean 1 5, prs11 scope mean 1 5, prs11 total mean 1 5, space familiarity 1 5, stai6 mean 1 5, thermal comfort 1 5: Psychological questionnaire outcomes.

14. Metadata / QC

- Date, Date (Atmo/LYS), Date (UCM), Date (eye), Datetime, GPS HDOP, GPS alt, GPS hdg, GPS lat, GPS lon, GPS speed valid sample count, GVI match dist m, MAG hdg, ParticipantID, PhaseID, pct complete: Metadata is excluded from the parameter list.
- HRV 5min available, HRV 5min max gap (s), HRV 5min removed pct, HRV 5min segment end, HRV 5min segment start, HRV 5min valid rri count, HRV 5min valid span (s): Metadata is excluded from the parameter list.
- SVF Aggregation Method, SVF Frames In Window, SVF Selection Method, SVF Status: Metadata is excluded from the parameter list.
- age, bike frequency, bmi category who, bmi kg m2, clothing accessories, clothing footwear, clothing height raw, clothing lower body, clothing skin tone, clothing upper body, clothing weight raw, commute modes, daily walking, gender, has children, height m, nationality, own bike for experiment, participant id, regular physical activity, residence type, residential area, response index, site familiarity 1 10, vision correction, weight kg, years in brussels: Metadata is excluded from the parameter list.
- end time s, phase start offset s, segment: Metadata is excluded from the parameter list.

STEP 2 — Domain Grouping (16 original candidates + 10 sound-sensitivity candidates)

Thermal	Index	Acoustic	Radiation	Pollution	Light	Greenery
AIR temp AIR RH WIND AWS	HUMIDEX PET UTCI	SND dBA Eq2 additions: freq_prop_20_125, vehicle_prop, wind_prop, freq_prop_8000_20000, mechanisms_prop, octave_prop_125_0_hz, urban_manmade_neutral_prop, freq_prop_4000_8000, urban_manmade_positive_prop, dominant_frequency_hz	MRT SUN Gh SVF	pm1 avg pm2.5 avg pm10 avg	lux avg kelvin avg	GVI 50m

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STEP 3 — Equations + design VIF (7 equations)

Equation	Maximum VIF
Eq 1 (Primary Model): UTCI * Sound + MRT * Sound + solar irradiance + SVF (sky view) + PM2.5 + lux + kelvin (light) + GVI (greenery)	3.07
Eq 2 (Sound sensitivity): Eq 1 + `freq_prop_20_125` + `vehicle_prop` + `wind_prop` + `freq_prop_8000_20000` + `mechanisms_prop` + `octave_prop_125_0_hz` + `urban_manmade_neutral_prop` + `freq_prop_4000_8000` + `urban_manmade_positive_prop` + `dominant_frequency_hz`	3.09
Eq 3 (Heat sensitivity — air temp, RH, wind): air temp * Sound + RH * Sound + wind * Sound + MRT * Sound + solar irradiance + SVF (sky view) + PM2.5 + lux + kelvin (light) + GVI (greenery)	2.55
Eq 4 (Heat sensitivity — HUMIDEX): HUMIDEX * Sound + MRT * Sound + solar irradiance + SVF (sky view) + PM2.5 + lux + kelvin (light) + GVI (greenery)	2.50
Eq 5 (Heat sensitivity — PET): PET * Sound + MRT * Sound + solar irradiance + SVF (sky view) + PM2.5 + lux + kelvin (light) + GVI (greenery)	3.02
Eq 6 (PM size sensitivity — PM1): UTCI * Sound + MRT * Sound + solar irradiance + SVF (sky view) + PM1 + lux + kelvin (light) + GVI (greenery)	3.06
Eq 7 (PM size sensitivity — PM10): UTCI * Sound + MRT * Sound + solar irradiance + SVF (sky view) + PM10 + lux + kelvin (light) + GVI (greenery)	3.07

Eq2 adds 10 sound main effects; original interactions and phase adjustment remain. The full design has been checked for all 26 outcome-specific samples. `scripts/H4_multivariate.R` defines all seven equations; Step 5 fits Eq2 only when `H4_SPEC$fit_sound_sensitivity` is TRUE (default FALSE). Published fit artifacts remain from the last six-equation run until rerun.

Centering, scaling and interpretation of the seven equations

All seven equation designs use the same exposure-preparation rule in `h4_step4_exposure_prep()` in `scripts/H4_core.R`. For each continuous exposure, subtract its mean within each participant-phase bout, then divide the deviations by their sample SD across the prepared analysis data. This is within-bout centering followed by common-SD scaling, rather than ordinary whole-sample z-scoring or a separate z-score within every bout. Phase contrast indicators are not standardized.

```
Scaled exposure = (measurement - participant-phase mean) / SD of all available centered deviations for that exposure
```

The means and SDs are calculated before outcome-specific complete-case filtering. Consequently, a variable uses the same scale across equations, but its mean and SD need not remain exactly zero and one in every fitted subset. Missing values are retained as missing during scaling.

Temperature measurement	Participant-phase average	Centered temperature
25 deg C	23 deg C	+2 deg C
32 deg C	30 deg C	+2 deg C

Both observations represent the same deviation from their bout average, despite their different absolute temperatures. Dividing by a common SD changes units; subtracting the bout mean changes the exposure variation used to estimate

the association. Scaling does not make the data normally distributed or correct measurement error.

Thermal-sound interactions are products of the centered, scaled thermal and SND dBA columns. Their intended question is: **within a participant-phase, does the thermal-physiology association depend on sound being above or below that bout's usual level?** Main effects are conditional on the other interacting exposure being at its bout mean. Outcomes retain their measurement units, except the specified HRV outcomes receive a natural log.

This specification does not directly estimate how differences in absolute average temperature and sound between bouts relate to physiology. Phase fixed effects do not restore the removed participant-phase exposure means. If H4 is intended to estimate within-bout and between-bout exposure associations, those components require explicit specification; ordinary z-scoring alone does not separate them.

Reference	What it supports	Limit of the comparison
Enders and Tofighi (2007), Centering Predictor Variables in Cross-Sectional Multilevel Models: A New Look at an Old Issue	Group-mean centering and the interpretation of within-cluster and between-cluster effects.	General methodological support; not a replication of this participant-phase design.
Curran and Bauer (2011), The Disaggregation of Within-Person and Between-Person Effects in Longitudinal Models of Change	Separating fluctuations within a person from differences in average exposure between people.	The participant-phase centering unit and common-SD scaling used here are study specifications.
Loeys, Josephy and Dewitte (2018), More Precise Estimation of Lower-Level Interaction Effects in Multilevel Models	Equation 11 explicitly uses centered predictors and their product: center first, multiply second, matching the interaction construction here.	The paper also examines cluster means and cross-level interactions; the construction alone does not establish that omitting those components is appropriate here.

The common-SD-scaled coefficients are changes in outcome units per common SD of exposure deviations, not average person-specific standardized effects or correlations. The exact nested random-effects and residual-correlation specification is not validated merely by citing a centering paper.

Pending specification checks: confirm whether H4 targets within-bout fluctuations, between-bout exposure differences, or both; review the treatment of interaction components accordingly. Also check the complete-case filter, which currently requires all selected exposures, including Eq2-only sound fields, for other equations. This can exclude observations unnecessarily. These issues have not been resolved or tested by refitting models as part of this diagram update.